Experiment 8

AIM: Use Postman to test sample API

1. Objective

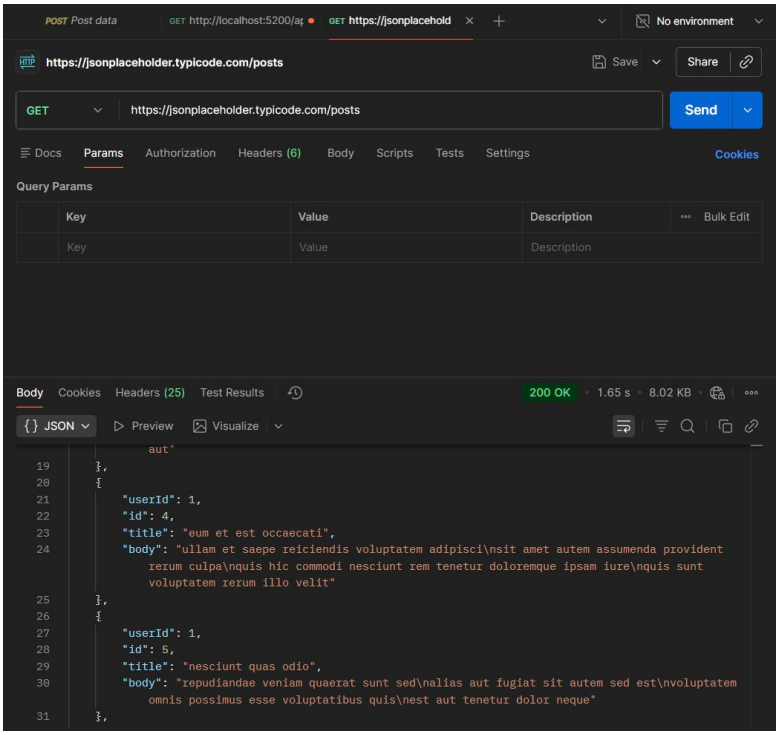
- Understand how to test REST APIs using Postman by sending different types of HTTP requests.
- Get
- Post
- Put
- Delete

2. Prerequisites

- postman
- API to be tested
    - Create a simple node.js API / use some free fake API

3. Test API

Get: URL: https://jsonplaceholder.typicode.com/posts



Post:

https://jsonplaceholder.typicode.com/posts

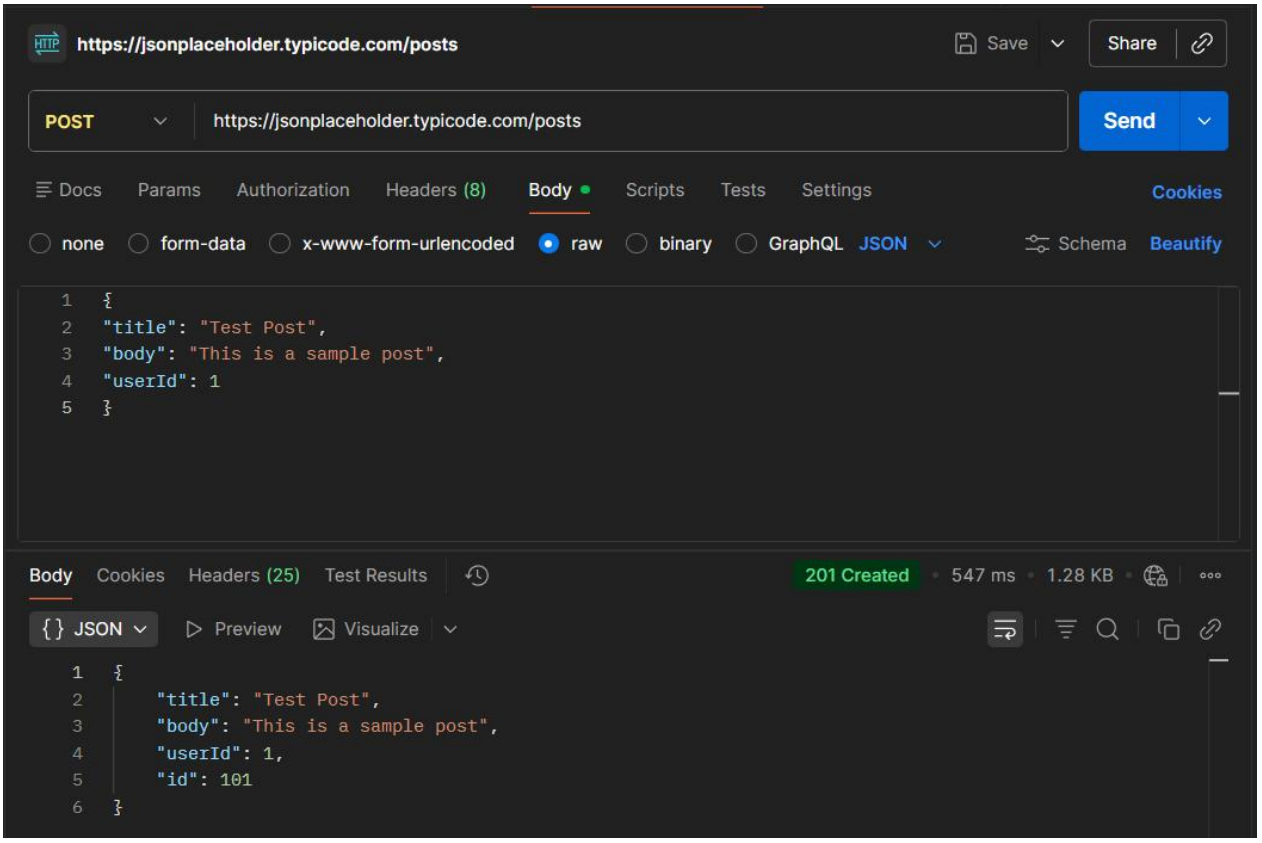
{

"title": "Test Post",

"body": "This is a sample post",

"userId": 1

}



Put:

https://jsonplaceholder.typicode.com/posts/1

{

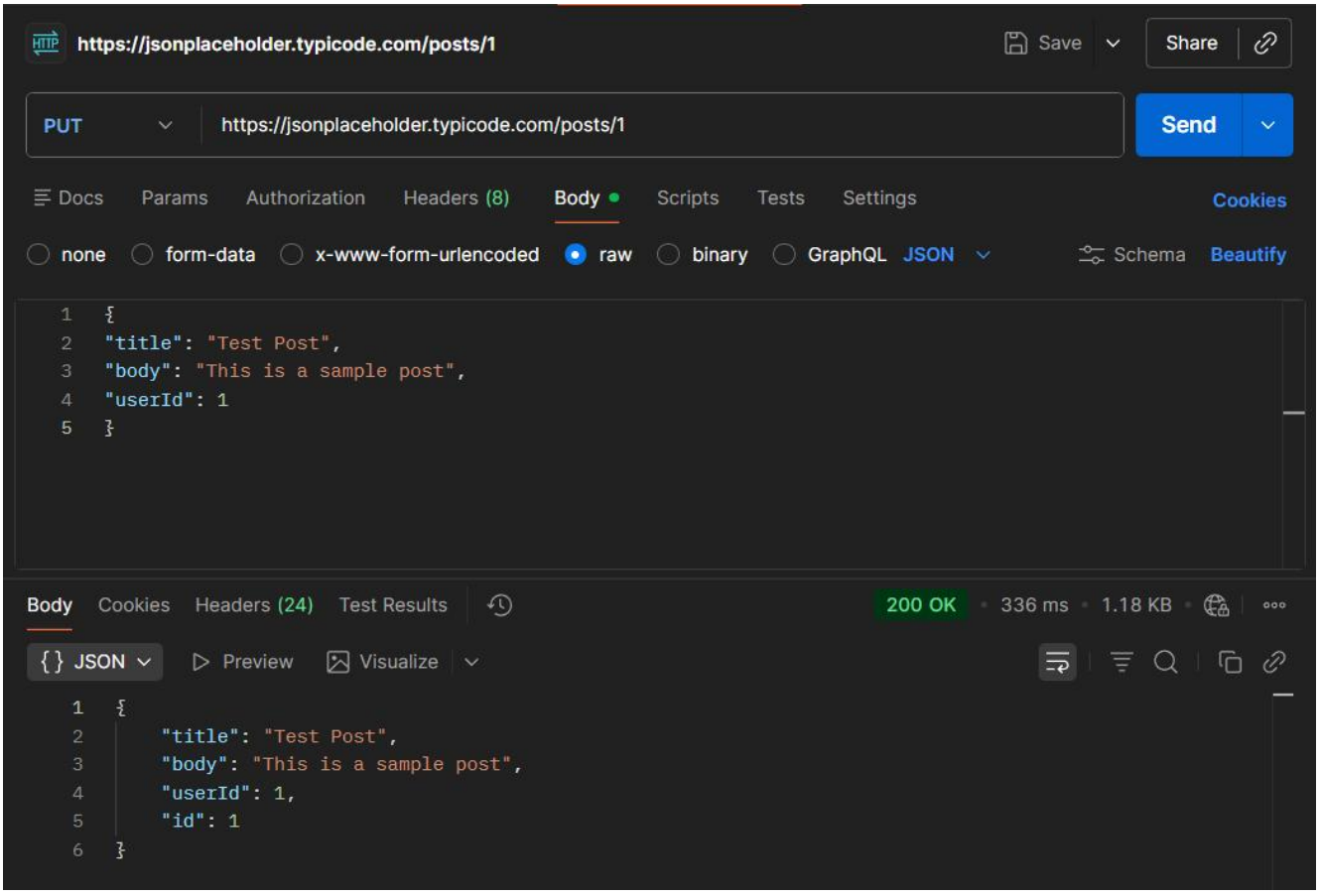
"id": 1,

"title": "Updated Title",

"body": "Updated content",

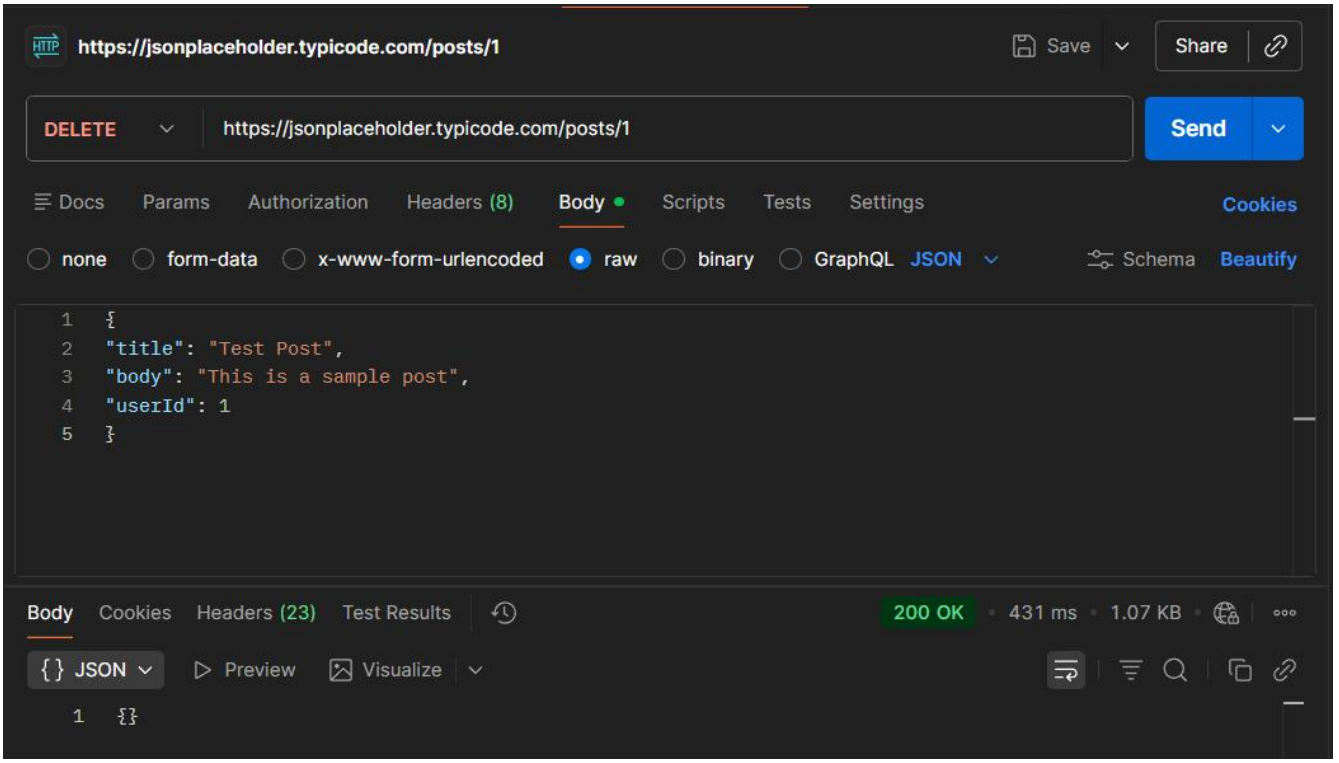
"userId": 1

}



Delete:

https://jsonplaceholder.typicode.com/posts/1



4. Create node.js API

Step 1: Create Project

mkdir simple-api

cd simple-api

npm init -y

npm install express

Step 2: Create server.js file

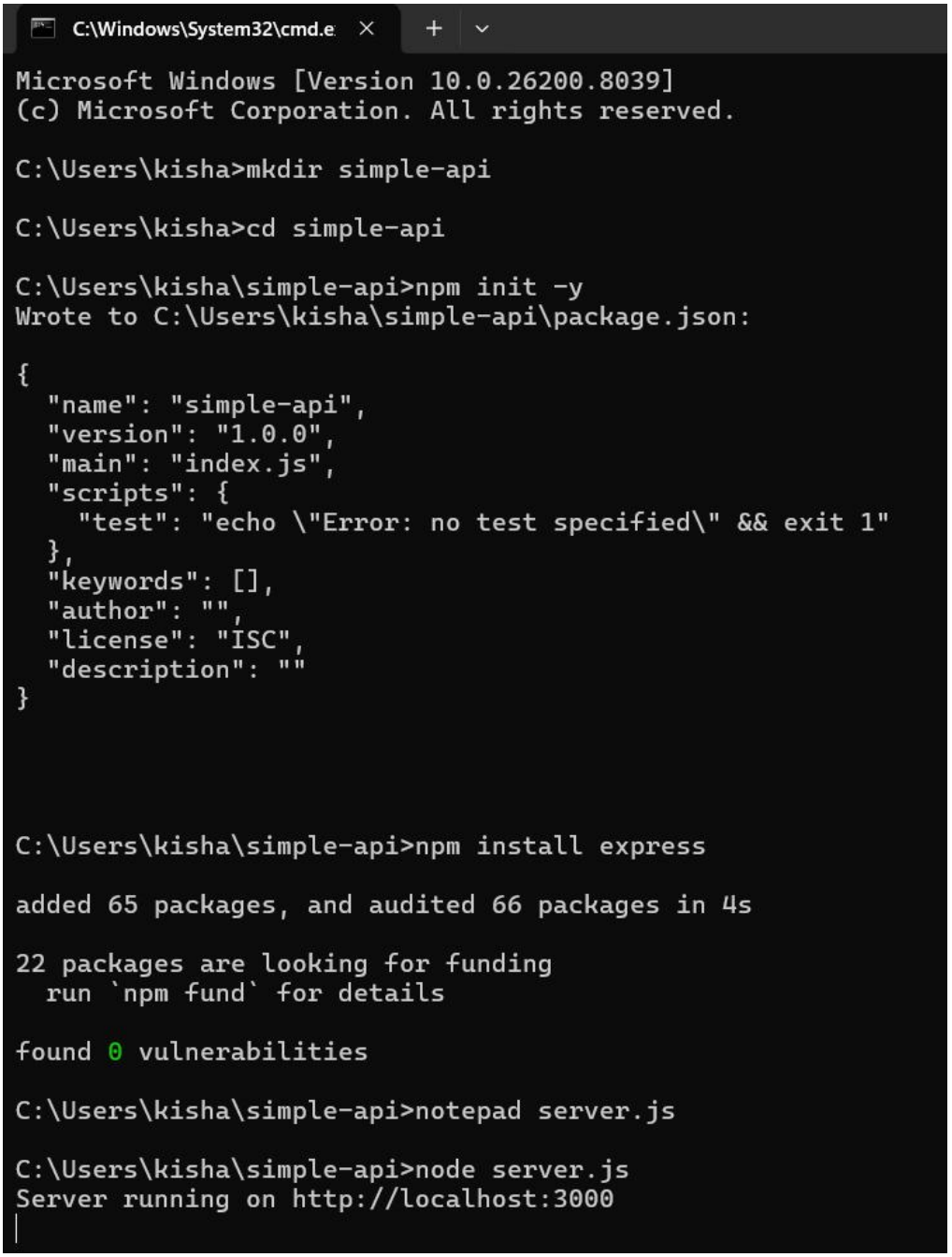
```
const express = require('express');
const app = express();
app.use(express.json());
let students = [
    { id: 1, name: "Kishan", branch: "CE" },
    { id: 2, name: "Tarak", branch: "CE" }
```

```
];
app.get('/students', (req, res) => {
    res.json(students);
});
app.get('/students/:id', (req, res) => {
    const student = students.find(s => s.id == req.params.id);
    res.json(student);
});
app.post('/students', (req, res) => {
    const newStudent = {
        id: students.length + 1,
        name: req.body.name,
        branch: req.body.branch
    };
    students.push(newStudent);
    res.status(201).json(newStudent);
    });
app.put('/students/:id', (req, res) => {
    const student = students.find(s => s.id == req.params.id);
    if (student) {
        student.name = req.body.name;
        student.branch = req.body.branch;
        res.json(student);
    } else {
        res.status(404).json({ message: "Student not found" });
    }
});
// DELETE - Remove student
```

```
app.delete('/students/:id', (req, res) => {
    students = students.filter(s => s.id != req.params.id);
    res.json({ message: "Student deleted" });
});
app.listen(3000, () => {
    console.log("Server running on http://localhost:3000");
});
```

Step 3: Run server

node server.js



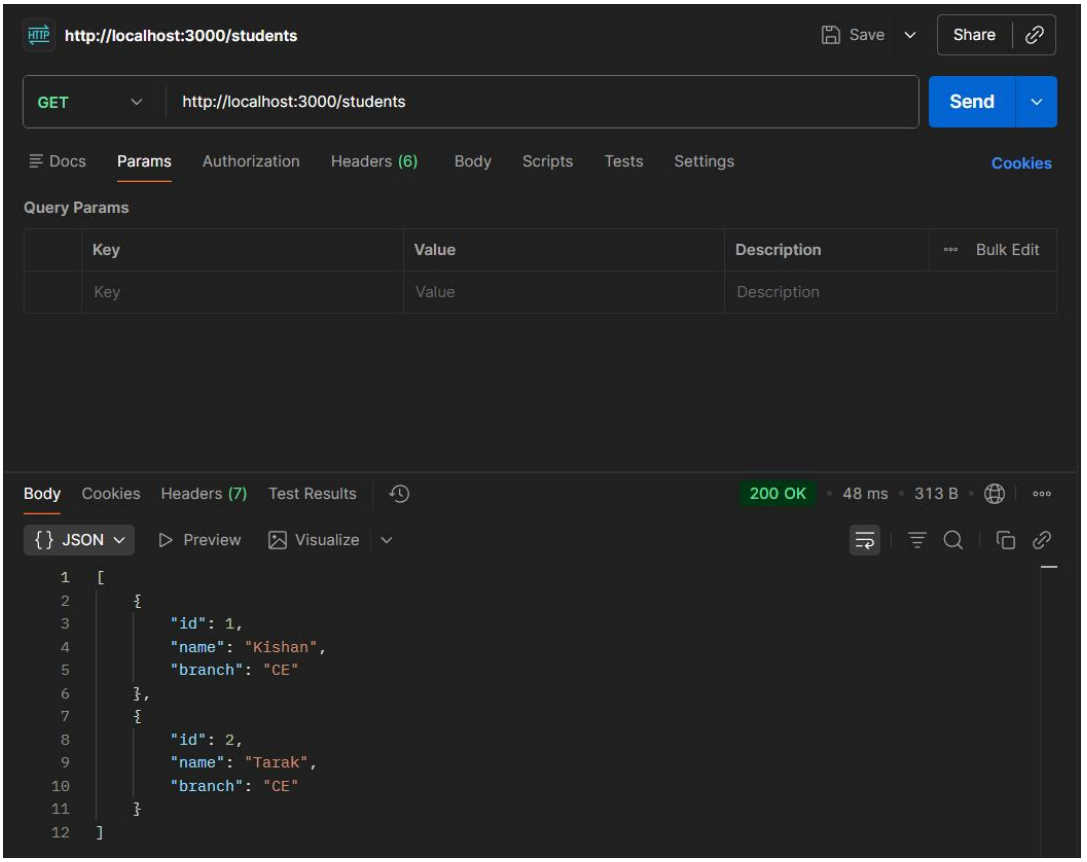
5. Procedure TEST using postman

1. GetAll:

Type: Get

URL : http://localhost:3000/students

Check response



2. Post:

Type: Post

URL: http://localhost:3000/students

Body:

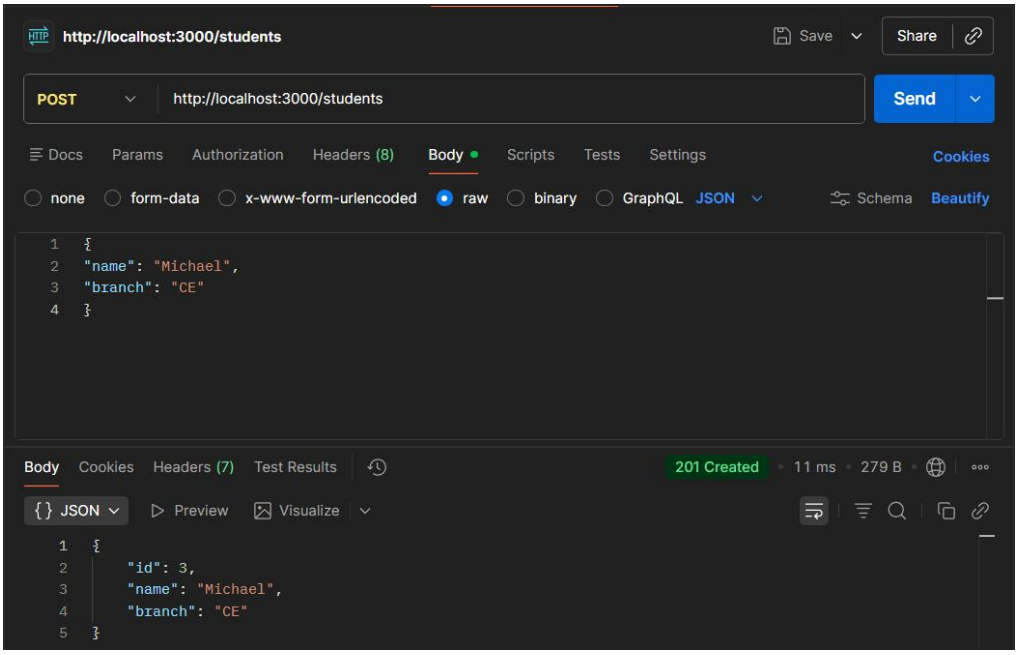
{

"name": "Michael",

"branch": "CE"

}

Check response and Send Get request to see new data inserted



3. Put request:

Type: PUT

URL : http://localhost:3000/students/1

Body:

{

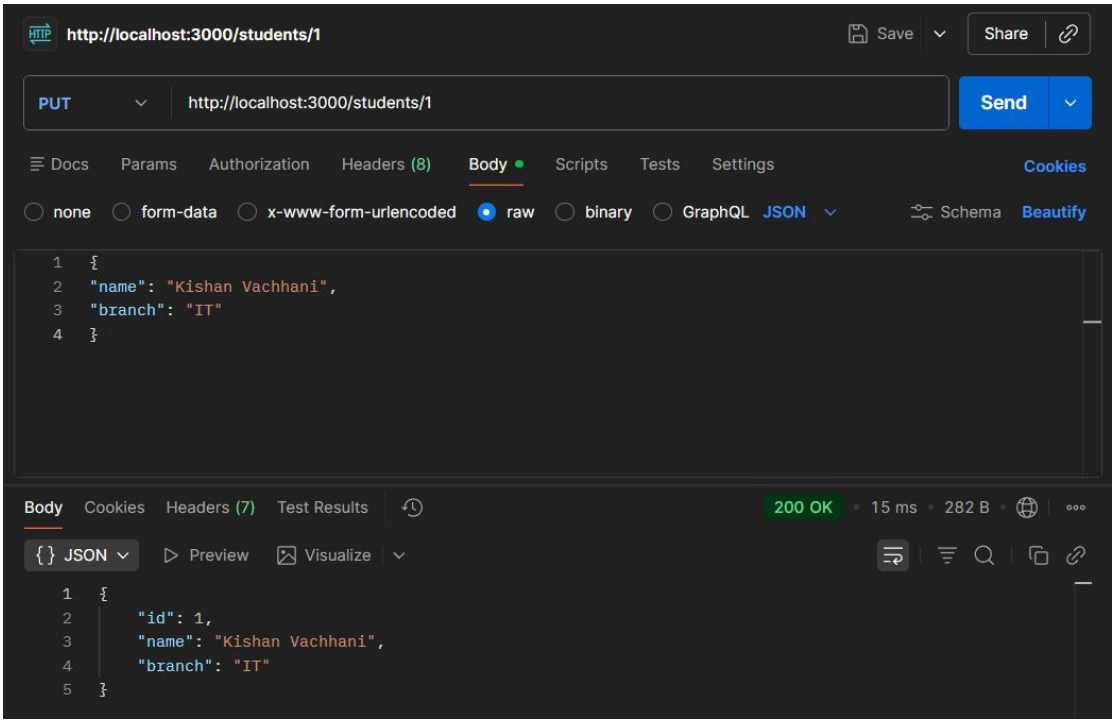
"name": "Kishan Vachhani",

"branch": "IT"

}

Check response

Send Get request to see updated data



4. Delete request:

Type: DELETE

URL : http://localhost:3000/students/1

Check response

Send Get request to see updated data